Parkinson's Telemonitoring

Interpretable Machine Learning for Predicting Severity from Acoustic Biomarkers

15.095 Machine Learning Under a Modern Optimization Lens

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The Clinical Problem

The Challenge

Parkinson's Disease (PD) progression varies widely per patient. Current monitoring relies on **infrequent, costly in-clinic visits** to assess the UPDRS (Unified Parkinson's Disease Rating Scale) score.

The Solution

Vocal Biomarkers. Speech changes (tremor, breathiness) correlate with severity. Recordings are cheap, scalable, and non-invasive, enabling continuous home monitoring.

Project Goal: Accuracy vs. Interpretability

Objective

Predict **total_UPDRS** using voice data from the UCI Telemonitoring dataset (~5,875 recordings, 42 subjects).

The Conflict

Black Box (Random Forest) models are accurate but uninterpretable.

Our Approach

Use **Mixed-Integer Optimization (MIO)** to select the optimal feature set, then build **Optimal Regression Trees (ORT)** for interpretable rules.

Data Overview & Challenges

Dataset Features

- **16 Acoustic Measures:** Jitter, Shimmer, HNR, NHR, RPDE, DFA, PPE
- **3 Demographic Features:** Age, Sex, Test Time
- **Total Features:** 19

Train/Validation/Test Strategy

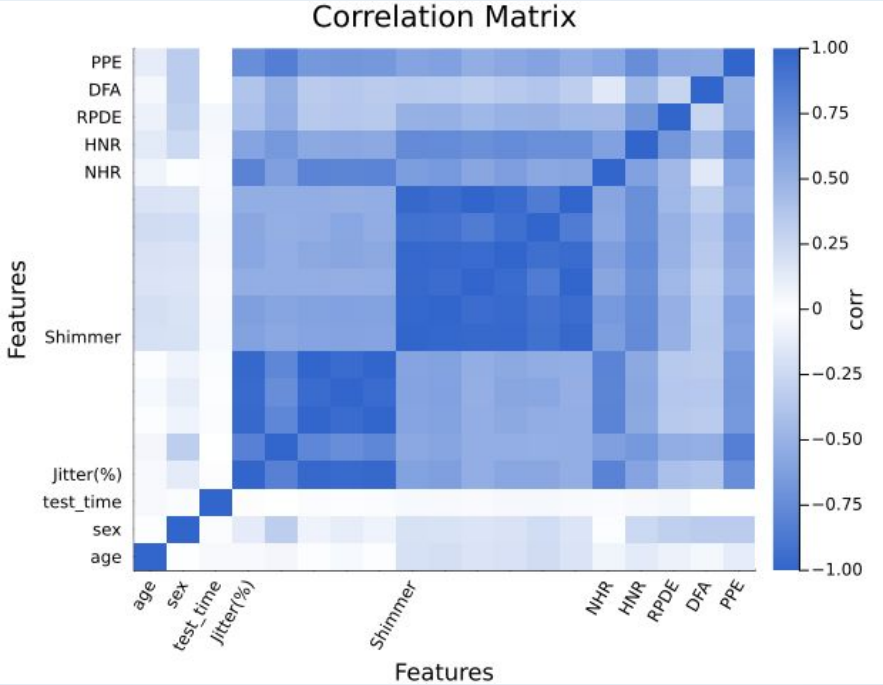
Subject Split

Ensures observations from the same subject stay together

Challenge: Multicollinearity

Acoustic features are highly correlated (higher than 0.8). Jitter and Shimmer variants measure similar properties.

Impact: Standard OLS regression is unstable. We need strong regularization and feature selection to isolate the true signal.



Phase 1: Feature Selection via MIO

Optimization Goal

We used MIO to act as a **Feature Selection Engine**. Unlike heuristic methods (Lasso), MIO solves for the **globally optimal** subset of k features. We tried several regression methods adding extensions iteratively, starting with Sparse Regression. Then we added robustness and ultimately performed Holistic Regression and compared the results.

Holistic Regression

- Sparsity
- Robustness
- Nonlinear Transformations
- Transformation Exclusivity
- Group Sparsity
- Multicollinearity Control
- Statistical Significance

Feature Sets

Demographics

Age

Voice Quality

HNR (Harmonics-to-Noise Ratio)

Shimmer:DDA_log

Jitter:DDP_log

Dynamics

PPE (Pitch period entropy)

Phase 2: From Holistic Regression to Tree Models

The Strategy

- **Input:** Best feature set from Phase 1 (holistic regression; small set of acoustic + demographic features)
- **Goal:** Turn these features into interpretable rules that predict UPDRS

Tree Models We Try

- **CART:** single tree, but greedy → can miss better splits
- **Random Forest / XGBoost:** more accurate but hundreds of trees → accuracy benchmark
- **ORT**: uses optimization to search for the best single shallow tree under depth limits

Tree Model Performance

ORT Test MAE

9.74

Improvement

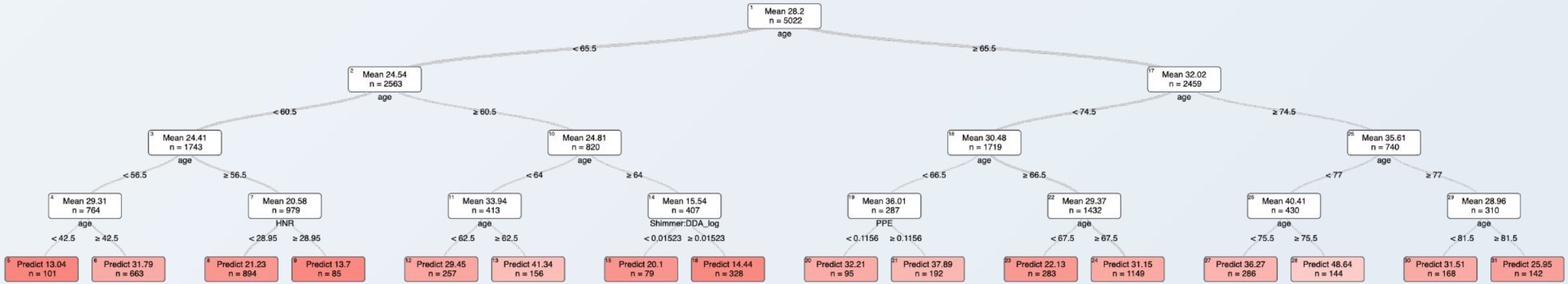
+7%

vs Random Forest

in Test RMSE

The ORT model outperformed the complex ensembles, likely due to the limited dataset size (42 subjects) where ensembles overfit.

Interpreting the ORT Results



- **Age is the primary driver:** younger (< ~60), middle-aged (60–70), older (> ~70); baseline UPDRS severity increases monotonically with age
- **Different biomarkers matter at different ages:**
  - **Younger patients (< 56):** Severity is mostly driven by voice quality (HNR)
  - **Older patients (> 66):** Severity depends more on pitch instability (PPE)
- **Subgroup-specific risk stratification:** e.g. older patients + high PPE → highest predicted UPDRS

Conclusion

Best Model: Holistic + ORT

Holistic feature engineering combined with ORT achieved the strongest performance, capturing nonlinear, age-dependent effects in disease severity

Interpretability with Accuracy

ORT provides strong predictive performance and maintaining transparency and interpretability as a single-tree model

Clinical & Real-World Impact

A small number of transparent rules using age and voice biomarkers enables scalable, low-cost remote monitoring of Parkinson’s disease